

# Seyed Abolfazl Mortazavi | Data Scientist | AI Engineer

Tehran, Iran

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## Professional Summary

Data Scientist and AI Engineer with hands-on experience building production machine-learning systems across speech, NLP, computer vision, and signal processing. At Shatel, own end-to-end data and ML workflows spanning collection, cleaning, analysis, training, evaluation, API delivery, and production monitoring. Experienced with large-scale speech and conversational datasets, statistical analysis, LLM-assisted text analytics, and communicating technical findings to product and business stakeholders.

## TECHNICAL SKILLS

- **Programming & Data Science:** Python, MATLAB, NumPy, Pandas, SciPy, Scikit-learn, XGBoost
- **Machine Learning & Deep Learning:** Classification, Regression, Feature Engineering, Model Evaluation, PyTorch, PyTorch Lightning, TensorFlow, Hugging Face Transformers
- **Statistical Analysis:** Exploratory Data Analysis (EDA), Hypothesis Testing (e.g., Chi-square and Kolmogorov–Smirnov tests), Statistical Model Evaluation
- **Speech & Audio AI:** Automatic Speech Recognition (ASR), Voice Activity Detection (VAD), Speech Emotion Recognition, Audio Preprocessing, Speech Enhancement, Audio Augmentation, Librosa, Speaker Identification, Speaker Verification
- **Natural Language Processing & LLMs:** Text Classification, Sentiment Analysis, Conversation Analysis, Text Summarization, Prompt Engineering, Transformer-based Models, End-of-Utterance Detection (EOT)
- **Computer Vision & Document AI:** Image Classification, Image Segmentation, Object Detection, Text Detection & Localization, Optical Character Recognition (OCR), Image Registration, Image Preprocessing & Enhancement, OpenCV
- **MLOps & AI Engineering:** MLflow, TensorBoard, FastAPI, Docker, Git/GitHub, Linux, Conda

## Professional Experience

### Shatel

Data Scientist / AI Researcher

Tehran, Iran

Jan. 2025–Present

- Own end-to-end data workflows for production AI systems involving speech, text, and document intelligence; processed and curated **hundreds of thousands of hours of speech data** and millions of conversational transcripts, with approximately **50,000 hours** used for model training.
- Developed and evaluated ASR systems achieving **2.97% WER** on evaluated Persian speech, **8.02% WER** in noisy call-center audio, and bilingual results of **4.02% Persian WER** and **2.06% English WER**.
- Independently designed and delivered a production call-center Quality Control and analytics system from data preparation through API integration; the system processes **1,000+ calls per day** and combines speech recognition, acoustic emotion analysis, transcript analytics, conversation-quality indicators, and automated summarization.
- Performed business and operational analysis on call-center data, identifying recording-quality, call-drop, and process issues; presented findings to stakeholders and informed corrective-action planning.
- Built a six-class emotion-recognition component for noisy call-center audio, achieving **87% Macro-F1**.
- Developed an end-of-utterance detection component for conversational systems, owning dataset preparation, labeling, modeling, evaluation, and integration while keeping proprietary model details confidential.
- Built preprocessing, text-detection, evaluation, and FastAPI layers for a multilingual OCR system covering Persian, Arabic, and English documents; achieved **97% F1** on text localization.
- Developed and evaluated a **16-bit speech-watermarking** pipeline for synthesized speech, achieving **0 BER on clean audio** and **<1% average BER** across evaluated perturbations including cropping, format conversion, noise, resampling, compression, speed changes, and pitch modification.
- Prepared technical reports and presented AI-system results and product capabilities in technical, product, and presales discussions.

### Independent Contractor

Freelance AI & Signal Processing Engineer

Nov. 2023–Present

- Deliver machine-learning, signal-processing, and computer-vision solutions for private client projects under NDA, covering data preparation, analysis, predictive modeling, and deployment.
- Translate client requirements into practical Python/MATLAB-based AI workflows while preserving confidentiality of client data and implementation details.

### Computational Neuroscience Lab, Amirkabir University of Technology

Research Assistant & Thesis Mentor

Tehran, Iran

Apr. 2025–Present

- Currently mentor six students (1 B.Sc., 4 M.Sc., 1 Ph.D.) on computational neuroscience, machine learning, signal processing, data analysis, and research methodology.
- Previously mentored two completed B.Sc. theses that received final grades of **20/20** and **19/20**.

#### Computational Neuroscience Lab, Amirkabir University of Technology

BCI Researcher

Tehran, Iran

Aug. 2022–Oct. 2023

- Developed quantitative EEG signal-quality measures and validated research hypotheses using statistical analysis, including Chi-squared and Kolmogorov–Smirnov tests.

## Education

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#### Sharif University of Technology

M.Sc. Electrical Engineering – Bioelectric, GPA: **19.02/20**; ranked **1st/17**

Tehran, Iran

2023–2025

**Thesis:** Eye Tracking Using EEG Signals During Simultaneous Functional Magnetic Resonance Imaging.

**Supervisor:** Dr. Ali Ghazizadeh Ehsaei    **Thesis grade:** 20/20 (Excellent)

#### Amirkabir University of Technology

B.Sc. Biomedical Engineering – Bioelectric, GPA: **19.07/20**; ranked **1st/121**

Tehran, Iran

2019–2023

**Additional studies:** Concurrent double-major coursework in Electrical Engineering (Telecommunications), Amirkabir University of Technology; GPA **19.38/20**, ranked **1st/132**. The second degree track was discontinued after admission to the M.Sc. program, with approximately 20 credits remaining.

## Selected Projects

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**Speech Enhancement:** Developed DSP-based speech enhancement and noise-reduction algorithms in MATLAB. [GitHub](#)

**Image Classification:** Built a PyTorch computer-vision pipeline achieving **98.37% accuracy** on a custom image dataset. [GitHub](#)

**NLP:** Developed machine-learning approaches for fake-news classification. [GitHub](#)

## Teaching Experience

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**Fall 2025:** Teaching Assistant, Blind Source Separation – Amirkabir University of Technology

**Fall 2024:** Teaching Assistant, Machine Learning – Sharif University of Technology

**Fall 2024:** Teaching Assistant, Neural Modeling – Sharif University of Technology

**Fall 2022:** Teaching Assistant, Engineering Mathematics – Amirkabir University of Technology

## Professional Development

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Neuromatch Academy Deep Learning; Deep Learning & Large Language Models (Sharif OCW); Advanced Learning Algorithms (Coursera); Complete Neural Signal Processing and Analysis; EEG Signal Processing

## Languages

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**Persian:** Native

**Turkish:** Native

**English:** Professional working proficiency